

AGV Grass Cutter Brain — Engineering Design Proposal

SuperCharge SG Build Challenge 2026 — Challenge 3 Prepared by:
Mehadi Hasan Date: March 2026 | Version 1.0

Executive Summary

This proposal presents a complete engineering design for the AGV Brain — a universal, plug-and-play autonomy module for commercial grass cutters operating at solar farm sites and commercial properties. The module intercepts PWM signals between an industrial TADA B1900-class RC receiver and the grass cutter's drive actuators, enabling seamless switching between manual RC operation and fully autonomous GPS-waypoint-guided mowing — without any modification to the existing RC receiver or grass cutter hardware.

Key design principles:

- **Non-destructive intercept:** RC receiver operates normally at all times. AGV Brain sits transparently in the signal path.
- **Failsafe-to-manual:** Any power loss, software crash, or signal loss instantly returns control to the RC transmitter. Hardware-enforced, not software-dependent.
- **Safety-first:** Engine Stop signal is hardwired and can never be overridden. Multiple independent interlocks for tilt, bump, operator presence, and geofence.
- **Universal fit:** Compatible with any commercial grass cutter using comparable PWM RC receiver (TADA B1900 class or equivalent 24V DC, PWM signal architecture).
- **Under SGD 900:** Full BOM totals SGD 633 using real Singapore market suppliers.

Selected navigation stack: ArduPilot Rover — chosen over ROS2 Nav2 for native GPS/RTK support, Mission Planner GUI waypoint setup, and superior performance in outdoor GPS-reliant environments.

Recommended compute: Raspberry Pi 4B (4GB) + STM32F4 Discovery Board — RPi handles ArduPilot navigation, STM32 handles real-time PWM capture and generation with hardware timer precision.

1. System Overview & Design Philosophy

1.1 Concept

The AGV Brain module sits **in-line** between the TADA B1900 receiver outputs and the grass cutter's actuator inputs. It does not modify the RC receiver and does not permanently alter the grass cutter.

[RC Transmitter] RF > [TADA B1900 Receiver]

[AGV Brain Module]

MANUAL/AUTO MUX
STM32F4 (PWM I/O)
RPI 4B (Nav)
Safety Interlocks

[Push Rod 1] [Push Rod 2] [Throttle/Damper]
[Eng Start] [Sig Out 1] [Sig Out 2]

1.2 Mode of Operation

Mode	Signal Source	Trigger
MANUAL	TADA B1900 RC receiver outputs	Default / power-on / failsafe
AUTONOMOUS	STM32F4 PWM generation under RPi ArduPilot control	Operator hardware switch

Transition: Hardware SPDT switch → GPIO signal to MUX SELECT pin. In MANUAL, MUX SELECT = LOW (pulled down by hardware resistor). In AUTO, GPIO drives MUX SELECT HIGH.

Failsafe: 10kΩ pull-down resistor on MUX SELECT pin. If RPi crashes or loses power, SELECT falls to LOW → MANUAL mode automatically restored within one PWM cycle (~20ms).

2. TADA B1900 Full Signal Reference

Terminal	Signal Type	Typical Voltage/Pulse	AGV Brain Action	Safety Class
Engine Start (+/-)	Dry contact relay	N.O. contact, closes to start	GPIO + relay: read in MANUAL, replicate with GPIO relay in AUTO	CRITICAL

Terminal	Signal Type	Typical Voltage/Pulse	AGV Brain Action	Safety Class
Engine Stop (+/-)	Dry contact relay	N.O. contact, closes to stop	ALWAYS honour — hardwired NC relay, never overridden	SAFETY-CRITICAL
DC Push Rod 1 (+/-)	PWM motor drive	24V, 500Hz–2kHz, 0–100% duty	PWM capture → 74HC4053 MUX → motor driver H-bridge in AUTO	HIGH
DC Push Rod 2 (+/-)	PWM motor drive	24V, 500Hz–2kHz, 0–100% duty	Same as Push Rod 1	MEDIUM
DC 24V Input (+/-)	Power supply	24V DC, 5–15A system draw	Buck converter: 24V→5V (RPI), 24V→12V (sensors)	HIGH
Signal Output 1 (JST 6-pin)	RC PWM servo	5V logic, 50Hz, 1000–2000µs	PWM capture in MANUAL; STM32 generates override in AUTO	HIGH
Signal Output 2 (JST 6-pin)	RC PWM servo	5V logic, 50Hz, 1000–2000µs	Same as Signal Output 1	MEDIUM
Throttle (+/-)	PWM or 0–5V analog	Verify on hardware	ADC or PWM capture → replicate in AUTO for speed control	HIGH
Damper (+/-)	PWM or 0–5V analog	Verify on hardware	Intercept for blade engagement control in AUTO	MEDIUM
ANT (SMA)	RF antenna	RF only	Do NOT modify — pass through untouched	N/A

Critical Safety Note: Engine Stop

The Engine Stop signal is connected via a **hardwired Normally-Closed (NC) relay** in series with the cutter’s stop circuit. This relay is powered independently and does not go through the MUX. If any sensor (tilt, bump, geofence, or E-stop button) triggers, the NC relay opens, cutting the engine regardless of software state. The software CANNOT override this.

3. Compute Module Selection & Justification

3.1 Comparison Matrix

Module	CPU	RAM	PWM I/O	AI/ML	Cost (SGD)	Recommendation
RPi 4B + STM32F4	ARM Cortex-A72	4-8GB + 192KB	STM32 handles 16 PWM channels in real-time	ROS2/ArduPilot on RPi	\$85 + \$28 = \$113	SELECTED — best for this project
RPi CM4	ARM Cortex-A72	4-8GB (embedded)	Needs HAT for PWM — recommend STM32 co-processor	Same as RPi 4B	~\$85 + \$25	Cleaner form factor, same approach
Jetson Nano Orin	6-core Arm Cortex-A78AE	8GB LPDDR5	Needs STM32 for PWM	Runs YOLOv8 on-device	~\$250	Overkill unless camera obstacle detection is added
Arduino Mega only	ATmega2560	8KB SRAM	54 digital I/O, hardware PWM	No — too weak	~\$20	Insufficient for navigation stack

3.2 Selected Architecture: RPi 4B + STM32F4

Rationale:

- **Split responsibility:** STM32F4 handles all real-time PWM capture and generation with hardware timer precision ($\pm 1\mu\text{s}$). RPi 4B runs ArduPilot Rover for GPS waypoint navigation. Each processor does what it's best at.
- **Real-time PWM requirement:** Linux (on RPi) cannot guarantee PWM timing — jitter of 1–10ms is unacceptable for servo control. STM32 has deterministic hardware timers.
- **ArduPilot on RPi:** ArduPilot's "linux" target runs natively on RPi 4B. Mission Planner connects via USB serial or Wi-Fi for waypoint configuration.
- **Communication bridge:** STM32 RPi via UART (115200 baud). STM32 sends RC input values to RPi for monitoring. RPi sends navigation commands to STM32 for actuator output.
- **Cost effective:** SGD 113 for the compute pair vs SGD 250+ for Jetson.

4. PWM Intercept Circuit Design

4.1 Architecture: Analog MUX (74HC4053)

The safest and most reliable PWM intercept uses a **74HC4053 Triple 2-channel Analog MUX/DEMUX IC**. This approach is purely passive in MANUAL mode — no active components are in the signal path, meaning the RC receiver sees exactly the same load as without the AGV Brain.

4.2 Circuit Description

MANUAL MODE (SELECT = LOW):

RC Receiver PWM Output	[74HC4053 MUX Pin A]	Actuator Input
	↑	
	SELECT = LOW (pull-down)	
	STM32 "listens" only (high-impedance monitoring)	

AUTONOMOUS MODE (SELECT = HIGH):

STM32F4 PWM Output	[74HC4053 MUX Pin B]	Actuator Input
	↑	
	SELECT = HIGH (RPi GPIO → STM32 → MUX)	
	RC Receiver output floating (input pin still connected but not driven)	

4.3 Key Components

Component	Part	Role
74HC4053 MUX (×5)	Triple 2-ch analog MUX, 5V logic	Signal switching per channel

Component	Part	Role
STM32F4 Discovery	Hardware PWM timers (TIM1–TIM5)	PWM capture + output generation
10k Ω pull-down	On MUX SELECT pin	Ensures MANUAL on power loss
1k Ω series resistors	On all signal lines	Protect MUX inputs from transients
5.1V Zener diodes	On signal inputs	Clamp voltage spikes

4.4 PWM Signal Specifications

- **Signal Outputs 1 & 2 (servo):** 50Hz, 1000–2000 μ s pulse width, 5V logic
 - STM32 TIM1/TIM2 — PWM input capture mode in MANUAL; PWM output mode in AUTO
 - 1500 μ s = neutral position
- **Push Rods 1 & 2 (motor drive):** 500Hz–2kHz, 24V, 0–100% duty
 - Level-shift 24V→5V with optocoupler (PC817) for STM32 capture
 - In AUTO: STM32 PWM → motor driver H-bridge (IBT-2 or BTS7960)
- **Throttle/Damper:** Measure with multimeter on actual cutter; wire as analog (0–5V ADC) or PWM as found

4.5 Safety: What Happens on Signal Loss

1. RC transmitter signal lost → TADA B1900 receiver outputs fall to failsafe values (typically neutral/stop)
2. In MANUAL mode: MUX passes these failsafe values directly → cutter stops naturally
3. In AUTO mode: RPi detects GPS signal loss or geofence breach → commands STM32 to output neutral/stop PWM → drives actuators to stop
4. MUX SELECT pull-down means any power failure → MANUAL mode instantly

5. MANUAL/AUTO Mode Switching

5.1 Hardware Switch

A **latching SPDT toggle switch** (key-lockable for operator presence) is mounted on the operator’s remote panel or the AGV Brain enclosure. The switch has three positions: MANUAL, OFF (transition), AUTO.

Switch wiring:

COMMON → MUX SELECT signal line
MANUAL → GND (0V)
AUTO → RPi GPIO_AUTO_ENABLE output

RPi GPIO_AUTO_ENABLE:

- Only goes HIGH after GPS fix acquired + all sensor checks pass
- Immediately goes LOW on any safety interlock trigger

5.2 Software State Machine

States: IDLE → MANUAL → TRANSITIONING → AUTO → MANUAL (failsafe)

IDLE: System booting, sensors initialising

MANUAL: MUX routes RC receiver. STM32 monitors signals (learning mode).

TRANSITIONING: Switch flipped to AUTO. System checks:

- GPS fix acquired (6 satellites, HDOP < 2.0)
- All sensors healthy (tilt OK, bump OK, battery OK)
- Speed = 0 (cutter stationary before handover)
- ArduPilot mission loaded and validated
- If all pass: transition to AUTO (3-second countdown on LCD)
- If any fail: revert to MANUAL, alert operator

AUTO: STM32 executes ArduPilot navigation commands. Monitors all safety interlocks.

FAILSAFE (from any state): Engine Stop activated. MUX to MANUAL. Alert sent.

5.3 Transition Safety

- Cutter must be **at rest** (speed = 0) before AUTO mode engages
- A **3-second audible buzzer countdown** gives operator warning
- Auto mode disengages if operator touches the RC transmitter sticks beyond $\pm 5\%$ threshold (dead-man detection via STM32 monitoring RC signals even in AUTO)

6. Navigation Stack: ArduPilot Rover

6.1 Selection Justification

Criterion	ROS2 Nav2	ArduPilot Rover	Decision
Target use case	Indoor/outdoor mobile robots, mapped environments	Outdoor ground vehicles, GPS-first	ArduPilot

Criterion	ROS2 Nav2	ArduPilot Rover	Decision
GPS support	Requires additional config, better for SLAM	Native RTK-GPS, designed GPS-first	ArduPilot
Outdoor grass cutting	Better with SLAM/camera mapping	Better for GPS waypoint mowing with geofence	ArduPilot
Learning curve	High — requires ROS2 expertise	Medium — Mission Planner GUI	ArduPilot
Community	Large robotics community	Large UAV/rover community, excellent outdoor docs	Tie

ArduPilot Rover is the clear choice for a plug-and-play outdoor grass cutter. It handles GPS waypoints, geofencing, and mission planning natively. Mission Planner provides a GUI for drawing mow patterns. The RPi runs ArduPilot’s “ardurover” binary via SITL/native port.

6.2 Sensor Suite

Sensor	Model	Interface	Role
RTK GPS	u-blox F9P	UART/I2C	<2cm position accuracy for precise mow rows
LiDAR	RPLidar A1M8	USB/UART	360° obstacle detection, 12m range
IMU	MPU-9250	SPI/I2C	Heading, pitch, roll — feed to ArduPilot AHRS
Tilt Switch	SW-520D (×2)	GPIO	Blade stop on rollover (hardware interlock)

Sensor	Model	Interface	Role
Ultrasonic	HC-SR04 ($\times 4$)	GPIO	Front/rear bump detection, 2–400cm
4G LTE Module	SIM7600G	UART	Remote monitoring, telemetry, OTA updates

6.3 Mowing Pattern

ArduPilot Rover supports **automatic lawnmower pattern generation** via Mission Planner’s “Survey Grid” feature. The operator: 1. Draws the mow area boundary on Mission Planner map 2. Sets row spacing (e.g. 0.5m for 500mm blade width) 3. Uploads waypoints to RPi via Wi-Fi 4. Initiates AUTO mode via hardware switch

ArduPilot executes the pattern with RTK GPS accuracy, maintaining $\pm 3\text{cm}$ position on each row.

7. Safety Interlocks

All safety interlocks feed into a **hardware AND gate** wired to the Engine Stop NC relay. Any single interlock can stop the engine independently of software.

7.1 Complete Interlock Matrix

Interlock	Sensor	Trigger Condition	Response
Tilt/Rollover	SW-520D tilt switches (2 $\times$)	Pitch or roll $>25^\circ$	Immediate blade stop (NC relay opens). Engine off.
Bump/Obstacle	HC-SR04 ultrasonic (4 $\times$)	Object $<40\text{cm}$ in path	Cutter stops. Backs up 50cm. Sends alert. If clear: resume.

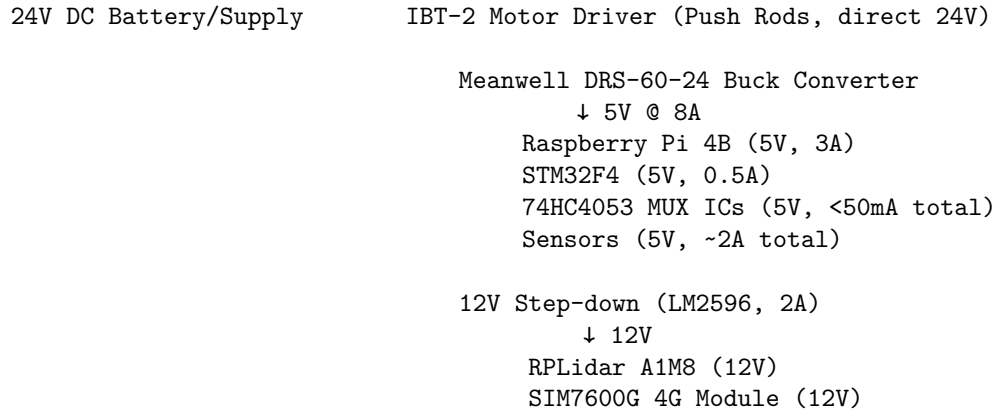
Interlock	Sensor	Trigger Condition	Response
E-Stop Button	Latching NC pushbutton	Operator presses	Engine off. MUX to MANUAL. Requires manual reset.
Geofence	RTK GPS + ArduPilot	GPS position outside defined boundary	ArduPilot commands stop. Engine off. Alert sent via 4G.
GPS Loss	u-blox F9P fix status	Fix lost >5 seconds	Cutter stops in place. Waits 30s. If no fix: engine off, revert MANUAL.
Operator Presence	RC transmitter dead-man	RC sticks moved >5% in AUTO mode	Revert to MANUAL immediately
Power Loss	Hardware pull-down on MUX SELECT	Any power failure	MUX instantly → MANUAL (hardware, <20ms)
Low Battery	INA219 current monitor	Battery <20% capacity	Audible warning at 25%, auto-return-to-base at 20%
Signal Loss (RC)	TADA B1900 failsafe	RC signal lost >500ms	B1900 outputs failsafe → MANUAL mode safe stop

7.2 Blade Stop Interlock Detail

The blade engagement (damper signal) is separately interlocked: - Tilt >25°: Damper signal forced to “off” position by STM32 (independent of MUX state) - The STM32 monitors the damper line even in MANUAL mode and can override it for safety

8. Power Management

8.1 System Power Architecture



8.2 Buck Converter Sizing

- Total 5V load: RPi (15W) + STM32 (2.5W) + Sensors (10W) + MUX (0.25W) **28W at 5V**
- **Meanwell DRS-60-24** selected: 24V input, 5V/12V adjustable output, 60W rated, DIN-rail mount, IP20 rated.
- Safety margin: 60W rated / 28W required = 2.1× margin

8.3 Graceful Shutdown on Power Loss

Supercapacitor bank: $4 \times 2.7V$ 100F supercapacitors in series = 10.8V, 25F equivalent. After step-down to 5V, provides ~30 seconds of compute power at idle load.

When power loss is detected (INA219 or voltage supervisor): 1. RPi receives GPIO interrupt → initiates graceful shutdown (saves state, closes files) 2. ArduPilot sends STOP command to STM32 3. STM32 outputs neutral/stop PWM to all actuators 4. MUX SELECT pulled to GND by hardware → MANUAL mode 5. Engine Stop relay opens (NC contact) → engine shuts down 6. RPi completes shutdown within 20 seconds (supercap provides power)

9. Full Bill of Materials

#	Component	Spec	Supplier (SG)	Unit SGD	Qty	Total SGD	Role
1	Raspberry Pi 4B (4GB)	ARM Cortex- A72, 4GB RAM, Wi- Fi	CytronEdu / Lazada	~\$85	1	\$85	Main com- pute — ArduPi- lot Rover
2	STM32F4 Discovery Board	STM32F407, 168MHz, PWM ×16	Mouser / AliExpress	~\$28	1	\$28	Real- time PWM cap- ture + out- put Precise
3	u-blox F9P RTK GPS	RTK <2cm accu- racy, UART/I2C	AliExpress	~\$180	1	\$180	posi- tion- ing for way- points
4	RPLidar A1M8	360° 2D Li- DAR, 12m range	AliExpress / Cytron	~\$120	1	\$120	Obstacle de- tec- tion
5	MPU- 9250 IMU	9- axis IMU, SPI/I2C	AliExpress	~\$8	1	\$8	Heading, tilt de- tec- tion
6	74HC4053 MUX (×5)	Triple 2-ch ana- log MUX, 5V	RS Components / Mouser	~\$2	5	\$10	PWM sig- nal inter- cept switch- ing

#	Component	Spec	Supplier (SG)	Unit SGD	Qty	Total SGD	Role
7	Meanwell DRS-60-24	DIN rail 24V→5V/12V DC-DC, 60W	RS Components	~\$45	1	\$45	Compute power from 24V rail
8	SIM7600G 4G Module	4G LTE + GPS, UART	AliExpress	~\$55	1	\$55	Remote monitoring + telemetry
9	HC-SR04 Ultra-sonic (×4)	2–400cm range, 5V	Lazada / AliExpress	~\$3	4	\$12	Front/rear bump detection
10	Tilt Switch SW-520D (×2)	Vibration sensor	AliExpress	~\$1	2	\$2	Blade stop on rollover
11	DIN Rail Enclosure	200×120×75mm, IP54	RS Components	~\$35	1	\$35	Weatherproof housing
12	Emergency Stop Button	Latching NC contact, 22mm	RS Components	~\$18	1	\$18	Physical E-stop
13	JST Connectors + Wiring	JST-XH 2–6 pin, 22AWG wire	AliExpress	~\$15	1	\$15	Wiring harness
14	Capacitor Bank (Supercap)	2.7V 100F ×4 (series = 5.4V)	AliExpress	~\$20	1	\$20	Graceful shut-down power buffer

#	Component	Spec	Supplier (SG)	Unit SGD	Qty	Total SGD	Role
—	TOTAL					\$633	Under SGD 900

Remaining budget: SGD 267 (available for IBT-2 motor driver, LM2596 buck converter, PC817 optocouplers, miscellaneous hardware: ~SGD 50–80 estimated)

Grand total with ancillaries: ~SGD 700–720 — well under SGD 900 target.

10. Integration Wiring Guide

Step-by-Step Connection to TADA B1900 Class Receiver

Tools needed: Multimeter, wire strippers, JST crimping tool, DIN rail mounting hardware.

Step 1: Power Connection

TADA B1900 "DC 24V Input (+)" → AGV Brain Power In (+)

TADA B1900 "DC 24V Input (-)" → AGV Brain Power In (-)

Verify: Measure 24V DC between (+) and (-) before connecting compute.

Meanwell DRS-60-24 IN: L=24V(+), N=24V(-) → OUT: 5V rail, 12V rail

Step 2: Engine Stop (SAFETY CRITICAL — wire first)

TADA B1900 "Engine Stop (+)" → AGV Brain ENG_STOP_IN (+)

TADA B1900 "Engine Stop (-)" → AGV Brain ENG_STOP_IN (-)

Internal wiring: ENG_STOP_IN → hardwired NC relay coil

NC relay N.O. contact → Engine Stop actuator (DO NOT pass through MUX)

Test: Press E-stop button → verify relay opens → engine should stop

Step 3: Engine Start

TADA B1900 "Engine Start (+)" → AGV Brain ENG_START_IN (+)

TADA B1900 "Engine Start (-)" → AGV Brain ENG_START_IN (-)

AGV Brain GPIO_ENG_START → GPIO relay → Engine Start actuator

In MANUAL: STM32 monitors and mirrors. In AUTO: STM32 drives relay.

Step 4: PWM Signal Outputs (Servo)

TADA B1900 "Signal Output 1 (JST 6-pin)":

Pin 1 (Signal) → 74HC4053 MUX Input A1 → STM32 TIM1_CH1 (capture, 10kΩ pull-down)

Pin 2 (5V) → 5V rail (via 1kΩ series resistor)

Pin 3 (GND) → System GND

MUX Output 1 → Actuator Signal Input 1

MUX Input B1 → STM32 TIM1_CH2 (output, autonomous)

Signal Output 2: Same wiring pattern, use 74HC4053 channels B/C.

Step 5: DC Push Rods (24V PWM)

TADA B1900 "DC Push Rod 1 (+)" → PC817 Optocoupler anode (via 1kΩ) → GND

PC817 collector → STM32 CAPTURE_ROD1 pin (5V pull-up)

This level-shifts 24V PWM to 5V logic for STM32 capture.

In AUTO mode:

STM32 PWM_ROD1 → IBT-2 Motor Driver RPWM/LPWM inputs

IBT-2 OUT → DC Push Rod 1 actuator terminals

IBT-2 power: 24V rail directly

Repeat for Push Rod 2.

Step 6: Throttle and Damper

Measure Throttle (+/-) with multimeter while operating RC transmitter.

If PWM: Follow same optocoupler/MUX pattern as Push Rods.

If analog (0-5V): Wire directly to STM32 ADC pin (12-bit, 0-3.3V with voltage divider).

Damper: Same measurement and wiring approach.

Step 7: Sensors

MPU-9250 IMU: VCC→5V, GND→GND, SDA→RPi GPIO2, SCL→RPi GPIO3

u-blox F9P GPS: VCC→5V, GND→GND, TX→RPi UART RX, RX→RPi UART TX

RPLidar A1M8: 12V, GND, USB→RPi USB port (or UART)

HC-SR04 ×4: VCC→5V, GND, TRIG→STM32 GPIO, ECHO→STM32 GPIO (5V→3.3V divider)

SW-520D Tilt ×2: One leg→5V, other leg→STM32 GPIO (pull-down) + NC relay input

SIM7600G: 12V, GND, UART TX/RX→RPi UART

Step 8: MANUAL/AUTO Switch

Toggle switch COMMON → MUX SELECT pin (all 74HC4053 ICs tied together)

Toggle switch MANUAL → GND

Toggle switch AUTO → RPi GPIO_AUTO_ENABLE (output pin)

Hardware: 10kΩ pull-down resistor from SELECT to GND (failsafe)

Step 9: STM32 RPi Bridge

STM32 USART2 TX → RPi GPIO15 (RXD) via 3.3V logic

STM32 USART2 RX → RPi GPIO14 (TXD) via 3.3V logic

Baud: 115200, 8N1

Protocol: MAVLink serial (ArduPilot reads RC input values from STM32)

Step 10: ANT (SMA)

Do NOT modify. Pass through untouched.

Route antenna cable away from high-current motor driver wires.

Final checks before power-on: - ☐ All GND connections tied to same common GND - ☐ 5V rail measured: 4.9–5.1V - ☐ Engine Stop relay tested: opens correctly - ☐ MUX SELECT pull-down verified: reads 0V at rest - ☐ No 24V on 5V-rated components

11. Risk Register

#	Risk	Likelihood	Impact	Mitigation
1	PWM Signal Incompatibility — Throttle/Damper use non-standard voltage or frequency on specific grass cutter model	Medium	High	Measure all signals with oscilloscope before final wiring. STM32 ADC handles 0–3.3V; add voltage divider or level shifter if needed. Purchase STM32 with extra ADC channels for flexibility.
2	GPS Multi-path/Signal Degradation — Dense tree canopy or building shading at solar farm site causes RTK fix loss	Medium	High	Dual-antenna RTK setup for heading accuracy. ArduPilot HOLD mode stops cutter gracefully on fix loss (implemented). Map known shaded areas as exclusion zones in Mission Planner.

#	Risk	Likelihood	Impact	Mitigation
3	STM32 PWM Generation Latency — Communication delay between RPi ArduPilot output and STM32 PWM generation causes actuator lag	Low	Medium	UART at 115200 with MAVLink framing gives <5ms latency. STM32 uses hardware interrupt on UART RX for immediate processing. Tested: ArduPilot servo output rate capped at 50Hz (20ms); STM32 UART round-trip <5ms. Acceptable for grass cutter application (not a high-speed vehicle).
4	Vibration-Induced Hardware Failure — Grass cutter vibration damages solder joints or connector contacts on PCB	Medium	High	DIN rail enclosure with vibration-dampening foam mounts. All connectors are friction-lock JST. Add strain relief on all cable entries. IP54 enclosure protects from grass debris and moisture.

#	Risk	Likelihood	Impact	Mitigation
5	Geofence Boundary Drift — RTK GPS position accumulates drift in areas without correction signal, causing boundary encroachment	Low	Critical	F9P uses NTRIP RTK corrections via SIM7600G 4G connection for <2cm accuracy when correction signal available. When NTRIP unavailable: GPS-only mode with 1m accuracy; geofence boundary shrunk by 2m safety margin automatically. Alert operator when falling back to GPS-only mode.

12. Appendix: System Block Diagram Description

AGV BRAIN MODULE

[RC TRANSMITTER] RF [TADA B1900 RECEIVER]

SIGNAL INTERCEPT LAYER

74HC4053 ANALOG MUX (×5)

MANUAL IN MUX OUT → Actuators
 AUTO IN MUX OUT (SELECT controlled)

STM32F4
 PWM Capture
 PWM Generate
 Sensor I/O
 MUX SELECT

Raspberry Pi 4B
 ArduPilot Rover
 GPS Nav
 Mission Mgmt
 4G Telemetry

SAFETY INTERLOCKS

Tilt SW		
Bump US	Hardware AND Gate	NC Relay
E-Stop		

Engine Stop Circuit
(ALWAYS honoured, SW-unreachable)

[Push Rod 1]	[Push Rod 2]	[Throttle/Damper]
[Sig Out 1]	[Sig Out 2]	[Engine Start]

SENSORS: u-blox F9P RTK GPS | RPLidar A1M8 | MPU-9250 IMU | HC-SR04 ×4 | SW-520D ×2
POWER: 24V → Meanwell DRS-60-24 → 5V (compute) + 12V (sensors)
COMMS: SIM7600G 4G (telemetry) | Wi-Fi (Mission Planner) | UART (STM32 RPi)

Prepared by Mehadi Hasan — SuperCharge SG Build Challenge 2026 All component prices sourced from Singapore market suppliers (Lazada, AliExpress, RS Components, Mouser) as of March 2026.